NETWORK SCANNING USING NMAP

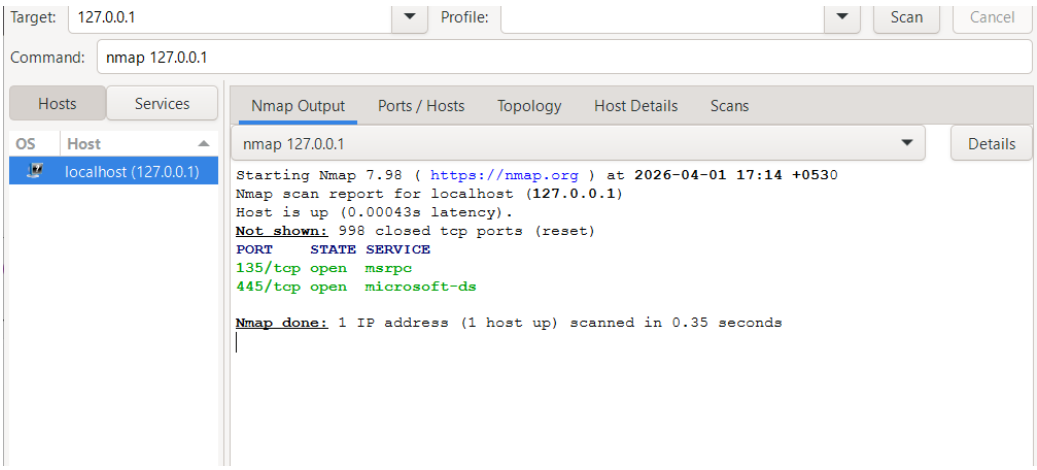
1. OBJECTIVE
   To perform network scanning using Nmap and identify open ports, running services, and system information to understand basic network security concepts.

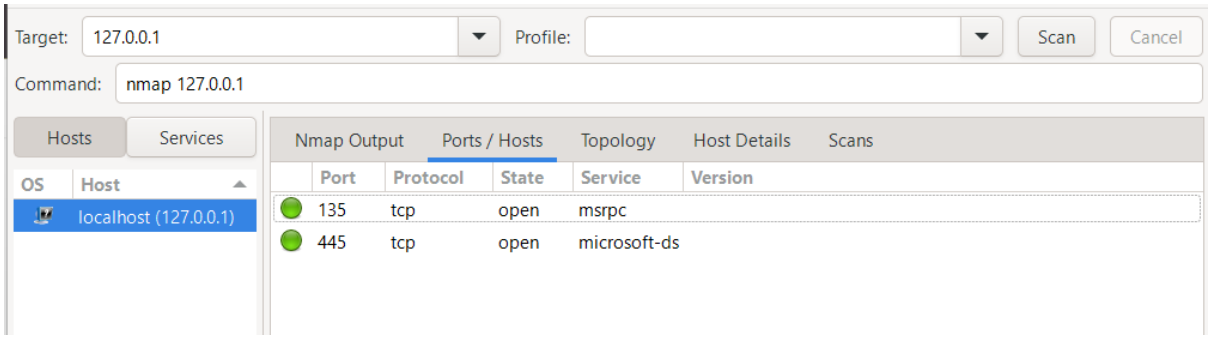
2. TOOLS USED
   Nmap

3. METHODOLOGY
   The following steps were performed during the experiment:
   - Installed and launched Nmap
   - Identified the system's IP address using ipconfig
   - Performed a basic scan on the local system (127.0.0.1)
   - Scanned the system using its IPv4 address
   - Performed service detection using -sV
   - Conducted an aggressive scan using -A for detailed analysis
   - Scanned the network gateway (router)
   - Analyzed the output to identify open ports and services

4. ANALYSIS

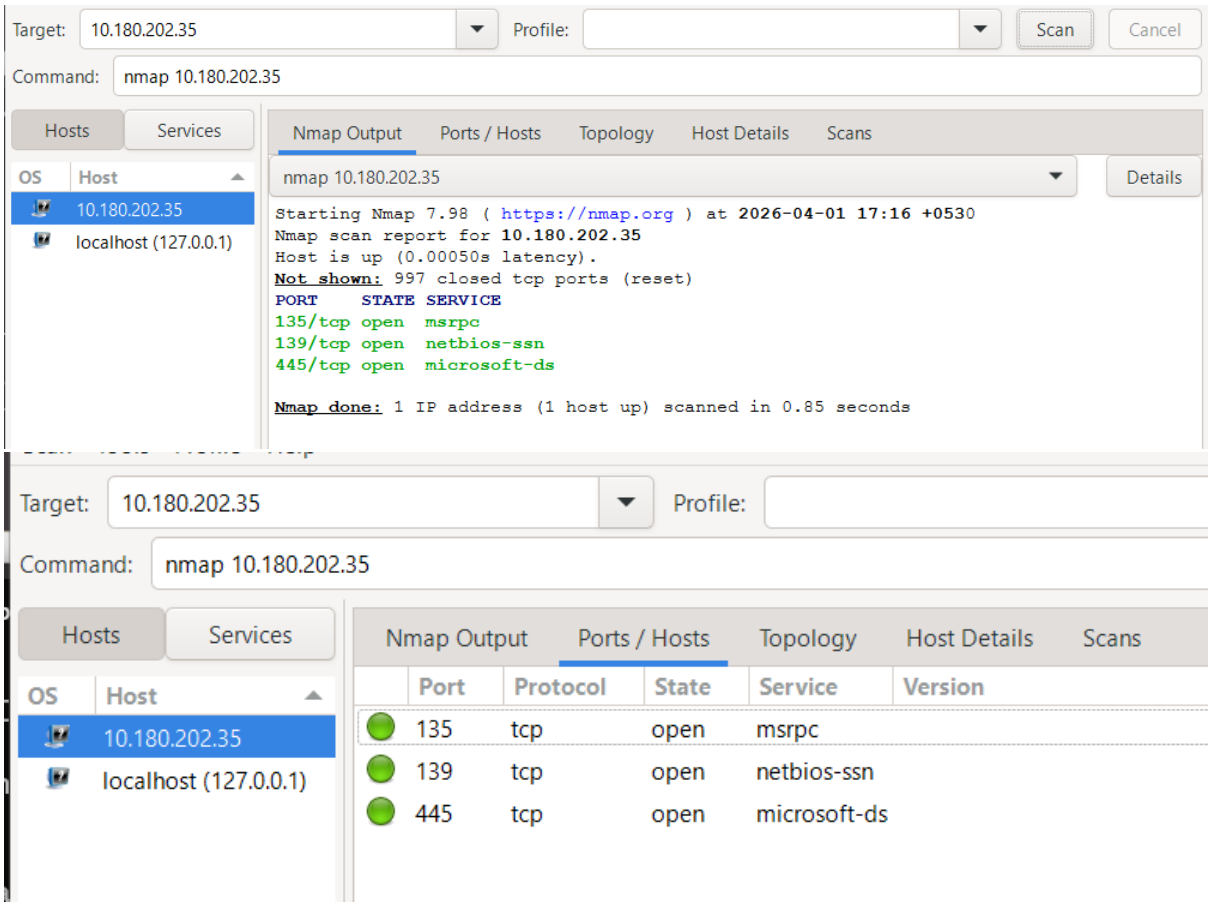
   4.1. Basic scan
   A basic scan was performed on the local system using the loopback address. The scan identified several open ports and associated services running on the system.





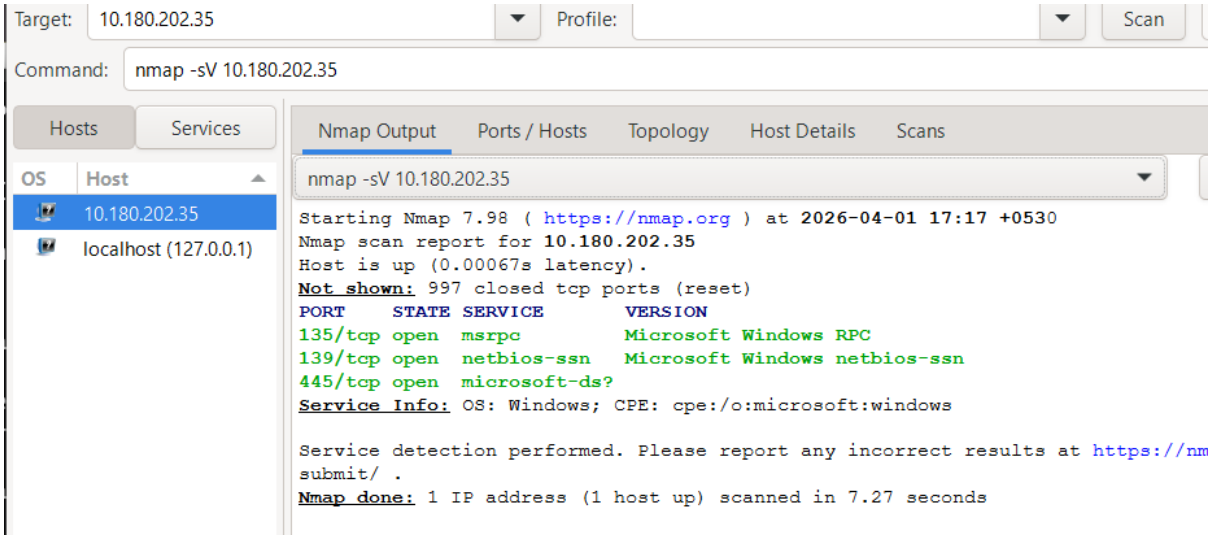
4.2. Scan using System IP Address

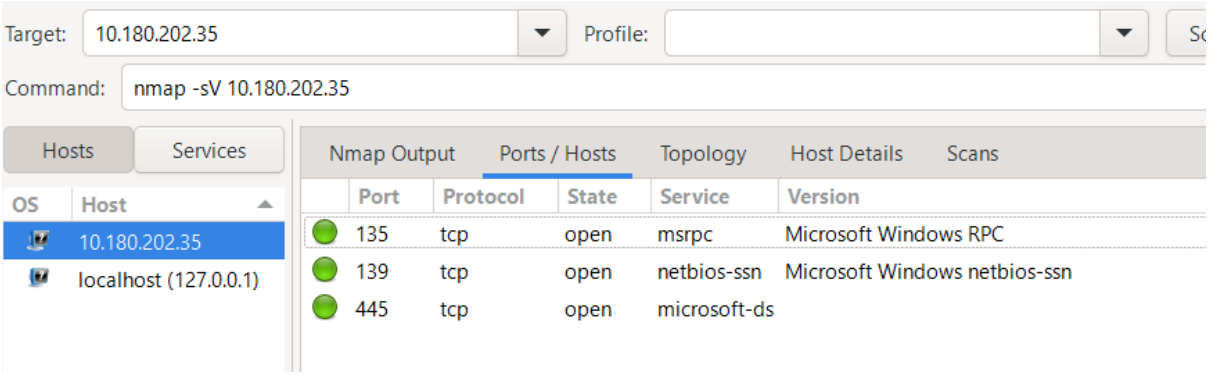
The system was scanned using its actual IP address. This simulates how the system appears to other devices on the network. Open ports were identified, indicating active services.



4.3. Service Version Detection

Service detection was performed to identify the versions of services running on open ports. This provides deeper insight into the system configuration and potential vulnerabilities.





4.4. Aggressive Scan

An aggressive scan was conducted which includes service detection, operating system detection, and script scanning. This provided detailed information about the system and running services.

nmap -A 10.180.202.35

Services

80.202.35

lhost (127.0.0.1)

Nmap Output | Ports / Hosts | Topology | Host Details | Scans

nmap -A 10.180.202.35

Details

```
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-01 17:19 +0530
Nmap scan report for 10.180.202.35
Host is up (0.00077s latency).
Not shown: 997 closed tcp ports (reset)
PORT     STATE SERVICE       VERSION
135/tcp open  msrpc         Microsoft Windows RPC
139/tcp open  netbios-ssn   Microsoft Windows netbios-ssn
445/tcp open  microsoft-ds?
No exact OS matches for host (If you know what OS is running on it, see https://
nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.98%E=4%D=4/1%OT=135%CT=1%CU=35010%PV=Y%DS=0%DC=L%G=Y%TM=69CD065
OS:4%P=i686-pc-windows-windows)SEQ(SP=101%GCD=1%ISR=107%TI=I%CI=I%II=I%SS=S
OS:%TS=A)SEQ(SP=101%GCD=2%ISR=10C%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=105%GCD=1
OS:%ISR=109%TI=I%CI=I%II=I%SS=S%TS=A)SEQ(SP=FC%GCD=1%ISR=10C%TI=I%CI=I%II=I
OS:%SS=S%TS=A)SEQ(SP=FD%GCD=1%ISR=103%TI=I%CI=I%II=I%SS=S%TS=A)OPS(O1=MFFD7
OS:NW8ST11%O2=MFFD7NW8ST11%O3=MFFD7NW8NNT11%O4=MFFD7NW8ST11%O5=MFFD7NW8ST11
OS:%O6=MFFD7ST11)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%W6=FFFF)ECN(R=
OS:Y%DF=Y%T=80%W=FFFF%O=MFFD7NW8NNS%CC=N%Q=)T1(R=Y%DF=Y%T=80%S=O%A=S+%F=AS%
OS:RD=0%Q=)T2(R=Y%DF=Y%T=80%W=0%S=Z%A=S%F=AR%O=%RD=0%Q=)T3(R=Y%DF=Y%T=80%W=
OS:0%S=Z%A=O%F=AR%O=%RD=0%Q=)T4(R=Y%DF=Y%T=80%W=0%S=A%A=O%F=R%O=%RD=0%Q=)T5
OS:(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=80%W=0%S=A%A=O
OS:%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T=80%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=
OS:N%T=80%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=80%
OS:CD=Z)

Network Distance: 0 hops
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled and required
| smb2-time:
|   date: 2026-04-01T11:49:29
|_  start_date: N/A

OS and Service detection performed. Please report any incorrect results at https://
nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 31.85 seconds
```

Nmap Output | Ports / Hosts | Topology | Host Details | Scans

10.180.202.35

Host Status

State:	up
Open ports:	3
Filtered ports:	0
Closed ports:	997
Scanned ports:	1000
Up time:	840471
Last boot:	Sun Mar 22 23:51:49 2026

Addresses

IPv4:	10.180.202.35
IPv6:	Not available
MAC:	Not available

Operating System

Name:	Microsoft Windows 10 1607 - 11 23H2
Accuracy:	

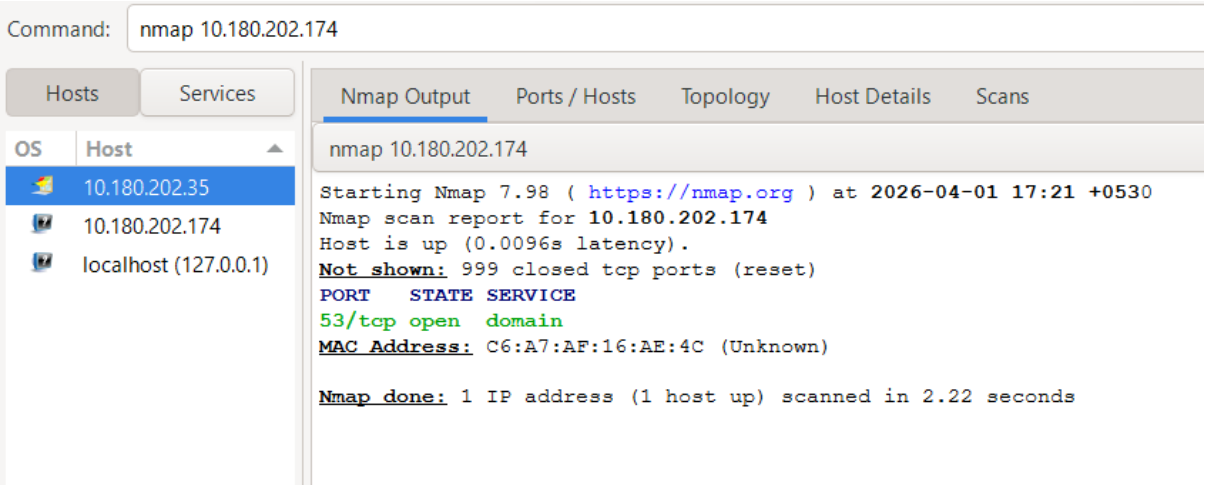
Ports used

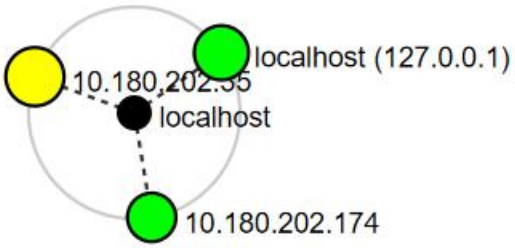
OS Classes

Type	Vendor	OS Family	OS Generation	Accuracy
general purpose	Microsoft	Windows	11	

4.5. Router Scan

The network gateway (router) was scanned using its IP address. The scan revealed open ports and services running on the router, demonstrating how network devices can also be analyzed using Nmap.





5. OBSERVATIONS

- Multiple open ports were identified on the system
- Each open port corresponds to a specific service
- Service version detection provides detailed system information
- The aggressive scan revealed additional system-level details
- The router scan showed that network devices also have open ports and services
- Open ports may represent potential security risks if not properly managed

6. CONCLUSION

This project demonstrated how Nmap can be used to perform network scanning and identify open ports, services, and system information. It highlights the importance of understanding network exposure and potential security risks in a system and network environment.